

Refresher: Statistics and Matrix Algebra

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Slide credits

- Keegan Korthauer

These are wonderful online books (hyperlinks):

- Modern Statistics for Modern Biology
- Introduction to Data Science
- Biomedical Data Science

Two challenges in genomic data analysis

- **Uncertainty**/noise distributions due to the nature of data-generating process
- **High-dimensional** and big data because of many variables and many samples
- The last lecture was to introduce genomics to statisticians

Today's goal is to introduce statistics to scientists

We need a general frameworks for addressing:

- ① How can we deal with the **uncertainty** of genomic data (statistics)?
- ② How can we effectively represent **high dimensional data** (matrix algebra)?

Note: In one lecture, it is impossible to teach you everything.

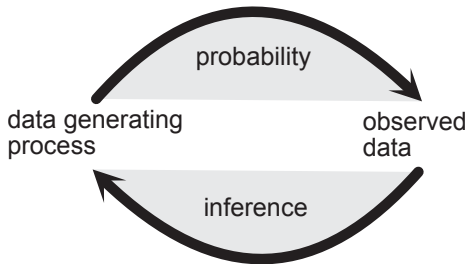
Today's lecture

- 1 Statistics Refresher
- 2 Matrix Algebra Primer
- 3 Matrix Algebra in Statistics

What is Statistics?

- The science of
 - **collecting, analyzing/modeling, interpreting** data
 - and **communicating uncertainty** about the results
- Statistics is **not** a collection of generic “recipes” to follow
- Our (statistical) goals for this course:
 - gain a rigorous understanding of core principles of common analyses of high-dimensional biological data
 - build solid foundation to follow up on specific topics

Probability and Inference



Wasserman, *All of Statistics* (2003)

- Probability will give insights into modelling.
 - Given a data generating process, we can we expect in data?
- We will estimate (recover) unknown from data by inference.
 - Given the data, what can we know about a plausible model?

$$X \sim P(\theta)$$

vs.

$$\hat{\theta} \leftarrow \arg \max \log P(\text{Data}|\theta)$$

A toy example: mutation in HIV genome

- Human Immunodeficiency Virus (HIV) Genome: 10^4 nucleotides
- Mutation rate: 5×10^{-4} per nucleotide
- How many mutations are expected to happen in each round of genome (RNA) replication?

See MSMB Chapter 1.

Expectation

- Suppose that a mutation can happen at any position with an equal probability.
- How many mutations are **expected** over all the genome per each replication?

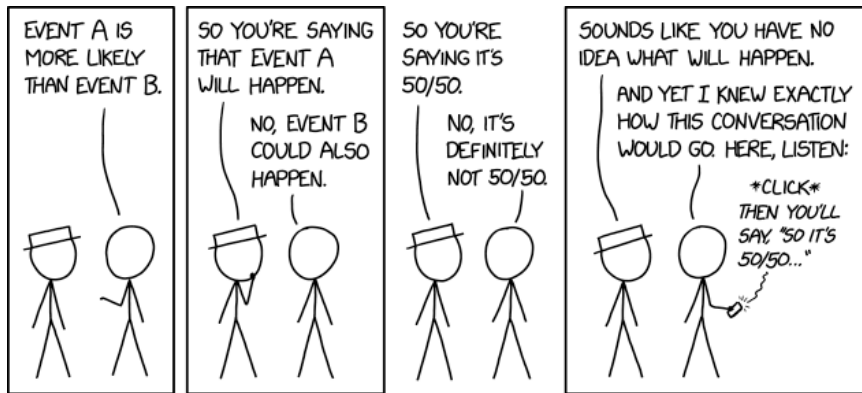
See MSMB Chapter 1.

A Bernoulli trial

We can have two possible outcomes (a variable) for each genomic location $i \in [10^4]$:

$$X_i = \begin{cases} 1 & \text{mutation occurred} & \text{with probability } 5 \times 10^{-4} \\ 0 & \text{no mutation} & \text{with probability } 1 - 5 \times 10^{-4} \end{cases}$$

So, will you see mutation or not?



XKCD on “prediction”

What do we call the mutation indicator variable X ?

- It's like a coin flipping with an unequal chance of head and tail.

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- We set a Bernoulli random variable $X = 1$, assigning the value 1 for an outcome of the mutation event (head), or $X = 0$ for the non-mutation (tail).

What do we call the mutation indicator variable X ?

- It's like a coin flipping with an unequal chance of head and tail.
- We call a variable like X : a **Random Variable**.
- We set a Bernoulli random variable $X = 1$, assigning the value 1 for an outcome of the mutation event (head), or $X = 0$ for the non-mutation (tail).
- It is a "random" variable, not a deterministic variable, because we assign the value with some probability.

Two types of random variables

- A **discrete** RV has a countable number of possible values
 - e.g. coin flip, throwing dice, genotype at a particular locus
- A **continuous** RV takes on values in an interval of numbers
 - e.g. blood glucose level, height of individuals

Probability of a Bernoulli trial

More succinctly, we can write the probability of the event:

$$P\{X_i = 1\} = 5 \times 10^{-4}$$

for all $i \in [10^4]$.

- More formally, probability maps an event to a long-run **frequency**.
- In order for a function to be a probability distribution/measure, it needs to satisfy the probability axioms.

Bernoulli distribution

Bernoulli distribution

We write

$$X \sim \text{Bernoulli}(p)$$

for a random variable X following Bernoulli distribution with the rate parameter p .

Bernoulli probability function

Then the corresponding probability function is

$$f(x) = p^x(1-p)^{1-x}$$

where $x \in \{0, 1\}$.

The lower case x is a particular value of the random variable X .

What is $f(x = 1)$? Or $f(x = 0)$?

- What is $\sum_{x \in \{0,1\}} f(x)$?

Each Bernoulli trial at each nucleotide

We have mutation rate

$$p = 5 \times 10^{-4},$$

with

$$n = 1$$

trial.

Let's see the relationship between Bernoulli and Binomial.

Bernoulli PMF:

$$f(x; p) = p^x (1 - p)^{1-x}$$

vs.

Binomial PMF:

$$f(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$$

if $x \in \{0, 1\}$ and $n = 1$?

Bernoulli is a special case of Binomial distribution

Binomial distribution

We write

$$X \sim \text{Binomial}(n, p)$$

for a random variable X counts the number of “success” events over total n trials with the success rate p .

Binomial probability function

Then the corresponding probability function is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

and

$$x \in \{0, 1, \dots, n\}.$$

Each Bernoulli trial at each nucleotide

We have mutation rate

$$p = 5 \times 10^{-4},$$

with

$$n = 1$$

trial.

$f(x; n, p)$ in R

```
size <- 1
prob <- 5 * 10^-4

## repeat experiments 10 times
rbinom(10, size = size, prob = prob)

## [1] 0 0 0 0 0 0 0 0 0 0
```

Are we seeing any mutation event?

Let's repeat Bernoulli trials many times

- In the HIV genome, what if we have $n = 100$ nucleotides?
- We still assume that each nucleotide can have mutation with $p = 5 \times 10^{-4}$.

```
size <- 100
prob <- 5 * 10^-4

## repeat experiments 10 times
rbinom(10, size = size, prob = prob)

## [1] 0 0 0 0 0 0 0 0 0 0
```

How about now?

Let's repeat Bernoulli trials many times

- What if we have $n = 10^4$ nucleotides?
- We still assume that each nucleotide can have mutation with $p = 5 \times 10^{-4}$.

```
size <- 10^4
prob <- 5 * 10^-4

## repeat experiments 10 times
rbinom(10, size = size, prob = prob)

## [1] 10 10 3 4 6 8 7 8 6 7
```

How about now?

Does rbinom generate the same value?

```
size <- 10^4
prob <- 5 * 10^-4

## repeat experiments 10 times
set.seed(1)
rbinom(10, size, prob)

## [1] 4 4 5 8 3 8 9 6 6 2
```

```
size <- 10^4
prob <- 5 * 10^-4

## repeat experiments 10 times
set.seed(2)
rbinom(10, size, prob)

## [1] 3 6 5 3 9 9 3 7 5 5
```


What R can do for us

If you type `?rbinom` or `help(rbinom)` for `Binomial(size,prob)`:

The Binomial Distribution

Description:

Density, distribution function, quantile function and random generation for the binomial distribution with parameters 'size' and 'prob'.

This is conventionally interpreted as the number of 'successes' in 'size' trials.

Usage:

```
dbinom(x, size, prob, log = FALSE)
pbinom(q, size, prob, lower.tail = TRUE, log.p = FALSE)
qbinom(p, size, prob, lower.tail = TRUE, log.p = FALSE)
rbinom(n, size, prob)
```

Functions related to probability

Probability Density Function (PDF)

Discrete $x \in \mathcal{X}$:

$$f(x) = P(X = x)$$

Continuous $x \in \mathcal{X}$:

$$\begin{aligned} f(x) &= dF(x)/dx \\ &= F'(x) \end{aligned}$$

Cumulative Distribution Function (CDF)

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= \sum_{t \leq x} f(t) \\ &= \int_{-\infty}^x f(t) dt \end{aligned}$$

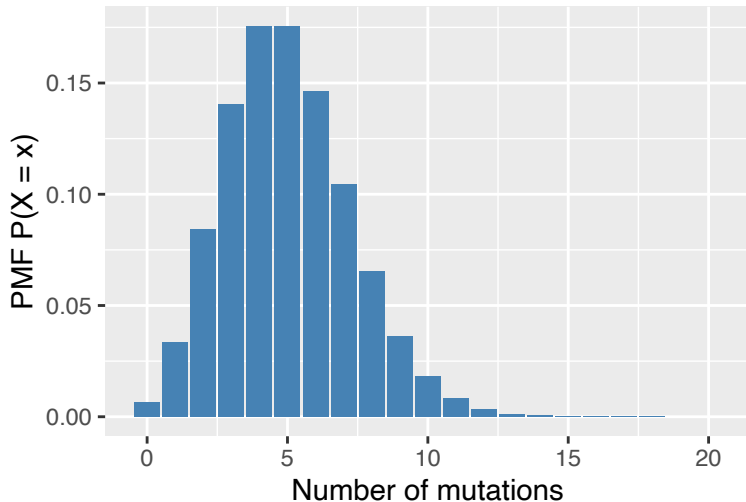
Quantile Function or inverse CDF

$$\begin{aligned} F^{-1}(p) &= \\ \inf\{x : F(x) \geq p\} \\ \text{for } p \in [0, 1] \end{aligned}$$

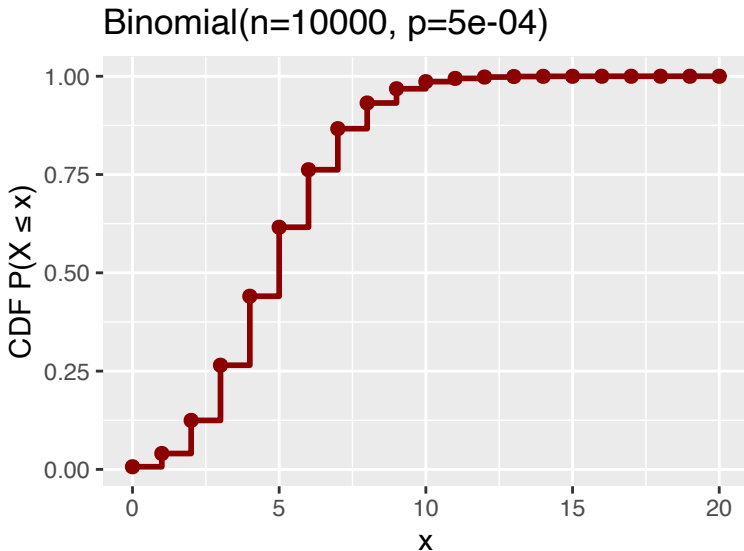
These definitions correspond to the R functions: `dbinom`, `pbinom`, `qbinom`.

Probability function dbinom

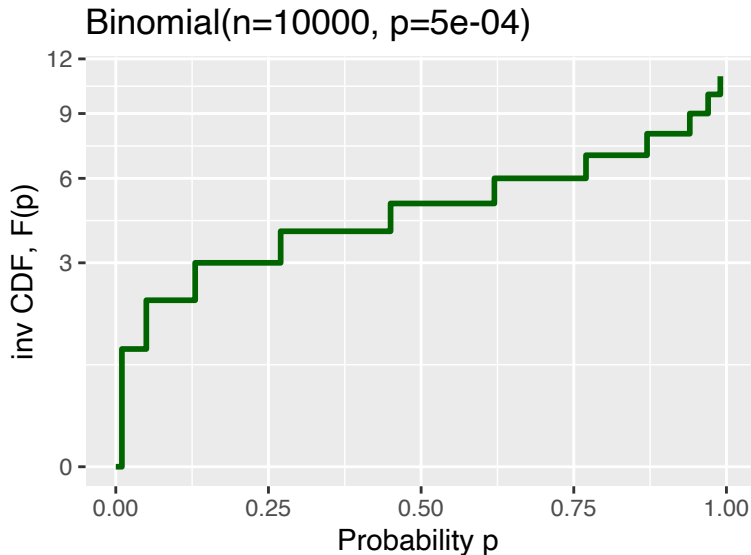
Binomial($n=10000$, $p=5e-04$)



Cumulative Distribution function pbinom



Quantile function qbinom



Let's discuss expectation (the 1st moment)

- What if we have $n = 10^4$ nucleotides?
- We still assume that each nucleotide can have mutation with $p = 5 \times 10^{-4}$.

Overall, what is the expected number of mutations?

$$X \sim \text{Binomial}(n, p)$$

Let's discuss expectation (the 1st moment)

- What if we have $n = 10^4$ nucleotides?
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Overall, what is the expected number of mutations?

$$\mathbb{E}[X] = n p$$

$$X \sim \text{Binomial}(n, p)$$

Let's discuss expectation (the 1st moment)

- What if we have $n = 10^4$ nucleotides?
- We still assume that each nucleotide can have mutation with $p = 5 \times 10^{-4}$.

Overall, what is the expected number of mutations?

$$\mathbb{E}[X] = n p = 10^4 \cdot 5 \times 10^{-4} = 5$$

$$X \sim \text{Binomial}(n, p)$$

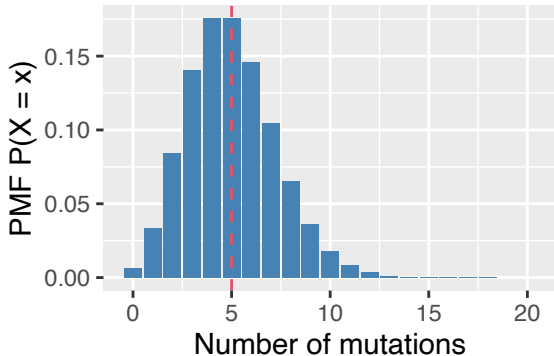
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$$X \sim \text{Binomial}(n, p)$$

Overall, what is the expected number of mutations?

$$\mathbb{E}[X] = np = 10^4 \cdot 5 \times 10^{-4} = 5$$



Formal definitions: Moments of a distribution

Expectation (μ)

Measures the average:

For discrete,

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot f(x)$$

For continuous,

$$\mathbb{E}[X] = \int_{x \in \mathcal{X}} x \cdot f(x) dx$$

Variance (σ^2)

Measures the spread:

$$\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2]$$

Equivalently:

$$\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Standard deviation:

$$\sigma = \sqrt{\mathbb{V}[X]}$$

Skewness

Measures asymmetry of the distribution:

$$\gamma_1 = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right]$$

Equivalently:

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

Moments of Binomial distribution

For $X \sim \text{Binomial}(n, p) \equiv \text{Binomial}(n = 10^4, p = 5 \cdot 10^{-4})$:

Expectation

$$\mathbb{E}[X] = np$$

Variance

$$\mathbb{V}[X] = np(1 - p)$$

Standard deviation:

$$\sigma = \sqrt{np(1 - p)}$$

Skewness

$$\gamma_1 = (1 - 2p) / (\sqrt{np(1 - p)})$$

```
mu <- n * p
sig2 <- n * p * (1 - p)
gam <- (1 - 2 * p) / sqrt(n * p * (1 - p))

c(mu, sig2, gam)
```

```
## [1] 5.0000000 4.9975000 0.4468781
```

What if the genome is very long, and the mutation events are very rare

For $X \sim \text{Binomial}(n, p)$, probability function:

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- Very long genome: $n \rightarrow \infty$
- Very rare mutation: $p \rightarrow 0$
- But finite number of mutations: $np = \lambda < \infty$

Poisson distribution for the number of mutations

For Poisson RV, $X \sim \text{Poisson}(\lambda)$, we have the distribution function:

$$f(x) \rightarrow \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

Moments:

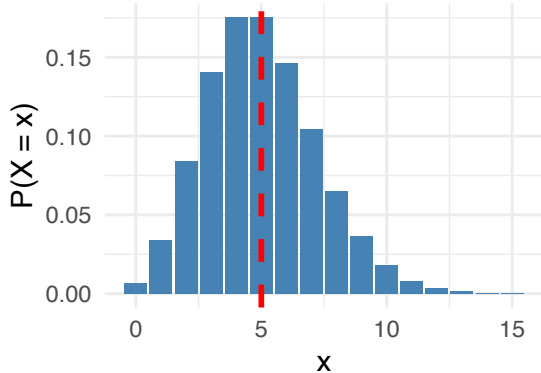
$$\mathbb{E}[X] = \lambda, \quad \mathbb{V}[X] = \lambda$$

Compare Binomial and Poisson

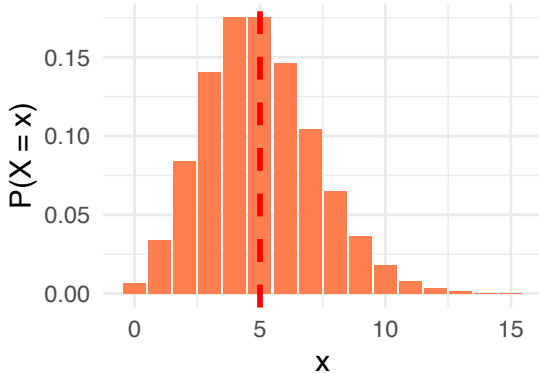
With the same parameters:

$$n = 10^4, p = 5 \times 10^{-4}, \lambda = np = 5$$

Binomial($n=10000$, $p=5e-04$)



Poisson($\lambda=5$)

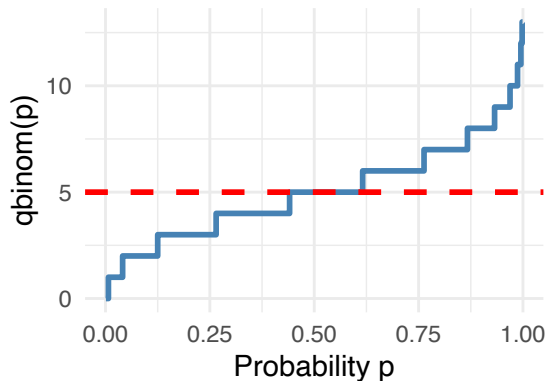


Compare Binomial and Poisson

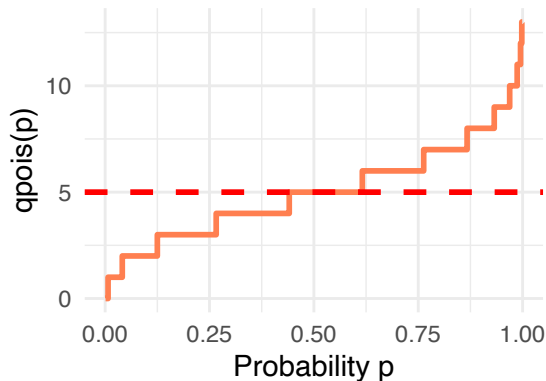
With the same parameters:

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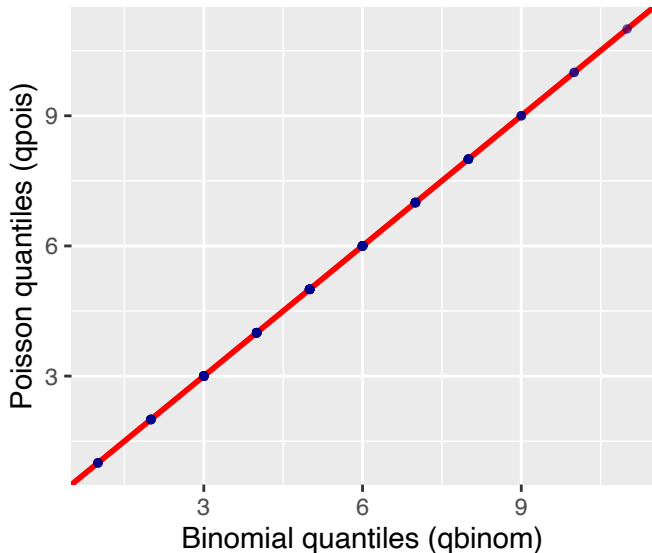
Binomial quantile function



Poisson quantile function



QQ-plot: Binomial vs Poisson



An example: $X \sim \text{Poisson}(\lambda = 5)$

Let's see what happens when we take sample means from $X \sim \text{Poisson}(5)$.

```
lambda <- 5  
xv <- rpois(10, lambda)  
print(xv)
```

```
## [1] 5 3 6 3 4 7 10 3 5 2
```

Statistical Inference is to figure out the unknown of a data generating process

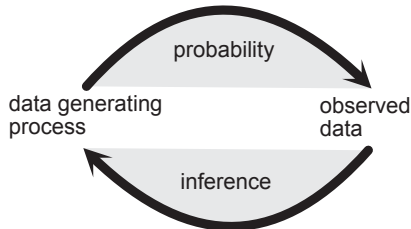
We knew (or assumed)

$$X \sim \text{Poisson}(\lambda)$$

and had some observed data set

$$D = \{ \\ 5, 3, 6, 3, 4, 7, 10, 3, 5, 2 \\ \dots \}$$

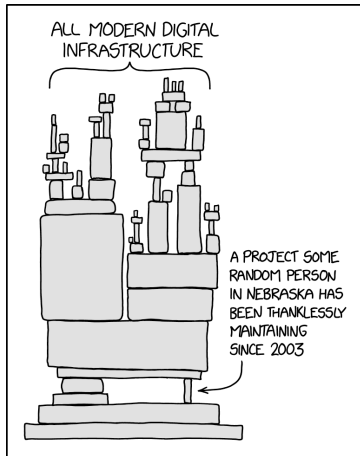
- Can we figure out the unknown λ ?



How can we estimate the unknown parameters?

- Assume $x_i \in D$ values are observation of independent and identically distributed (i.i.d.) X random variable.
- Goal: Estimate the unknown parameter (thus the corresponding data generating scheme) from observed *noisy* data...

Can we? Do we have some theoretical foundation?



<https://xkcd.com/2347/>

Can we estimate mean parameter by sample mean?

(Weak) Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be i.i.d. with the mean λ

The sample mean converges to the true mean λ :

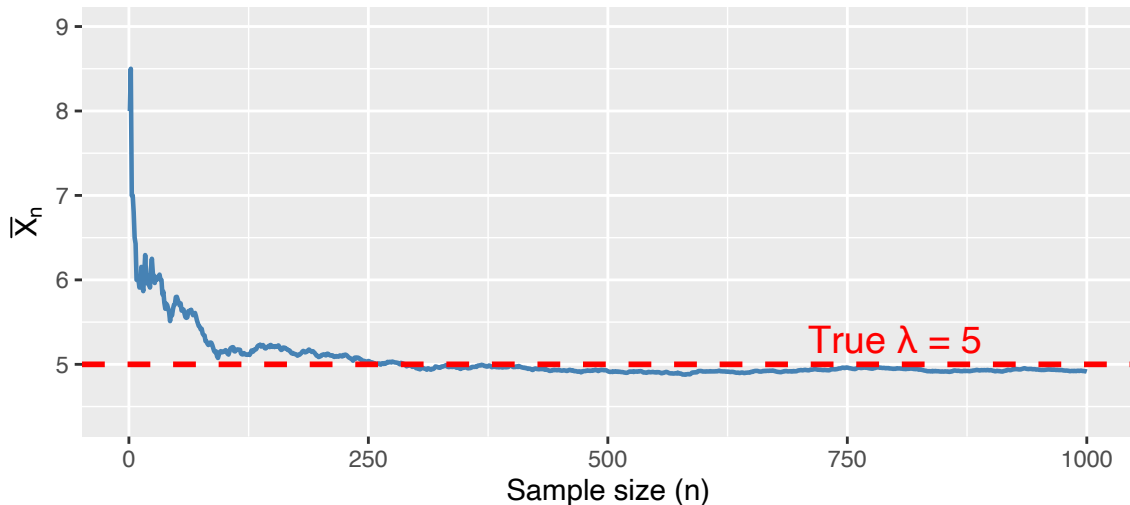
$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \lambda$$

as $n \rightarrow \infty$.

- With more observations, $n \rightarrow \infty$, the sample mean gets closer to the true mean
- $\xrightarrow{p}$: This is **convergence in probability**, meaning $P(|\bar{X}_n - \lambda| > \epsilon) \rightarrow 0$ for any $\epsilon > 0$
- Using sample mean to estimate mean is okay with high probability.

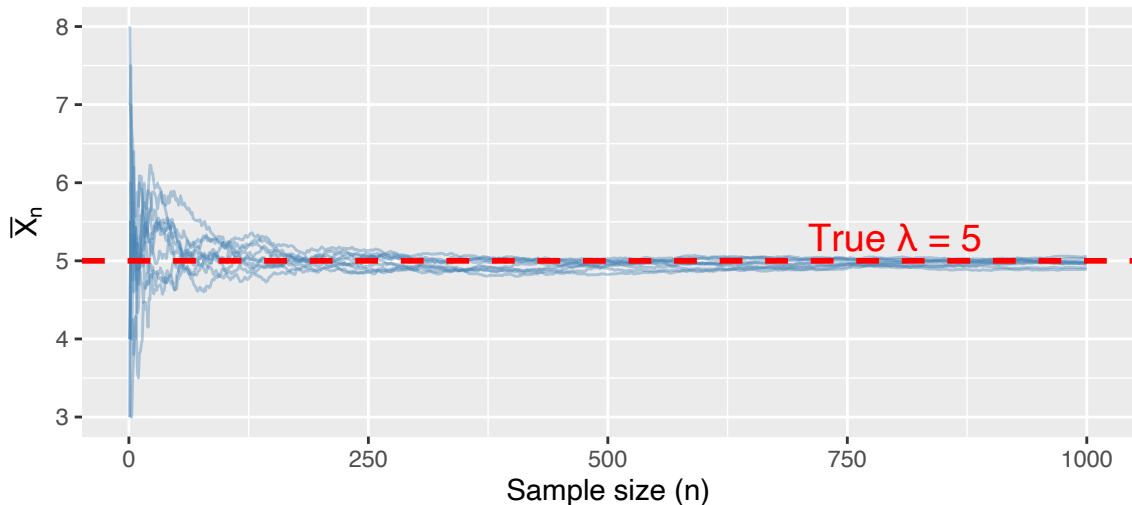
Example: Estimating λ from Poisson(5)

Sample mean converges to true mean



Multiple realizations of the LLN

10 independent realizations



Central Limit Theorem

Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean $\mathbb{E}[X_1] = \mu$ and variance $\mathbb{V}[X_1] = \sigma^2$.

The sample mean $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ converges in distribution to a normal distribution:

$$\bar{X}_n \xrightarrow{d} \mathcal{N}(\mu, \sigma^2/n)$$

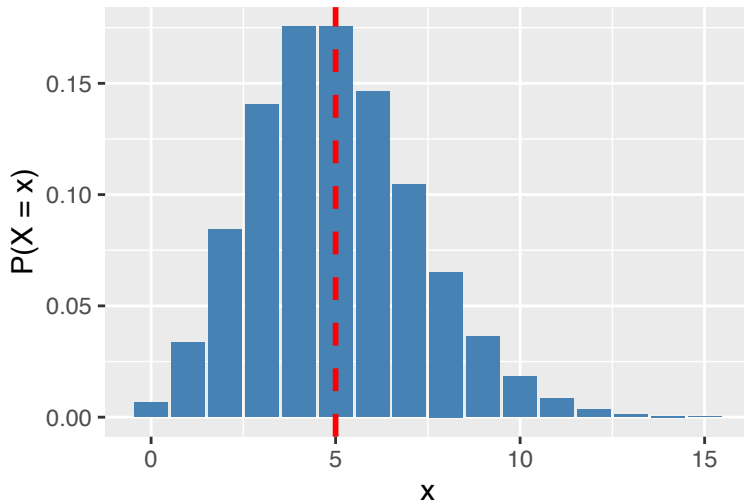
as $n \rightarrow \infty$.

- For any distribution, CLT holds!
- The distribution of sample mean $\bar{X}_n$ becomes approximately normal for large n
- The variance of $\bar{X}_n$ decreases as n increases (how fast?)

Well, let's if that really works empirically.

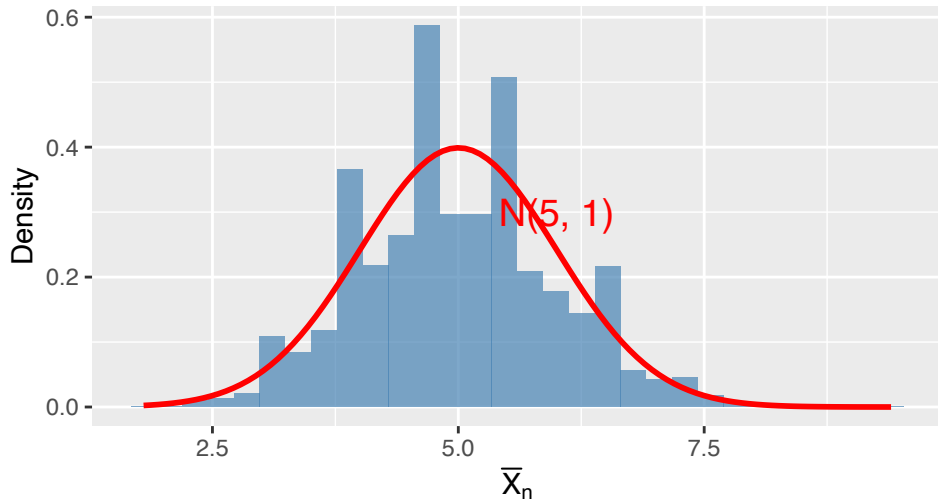
The sample Poisson example with data size $n = 1$

Poisson($\lambda=5$) - Individual observations



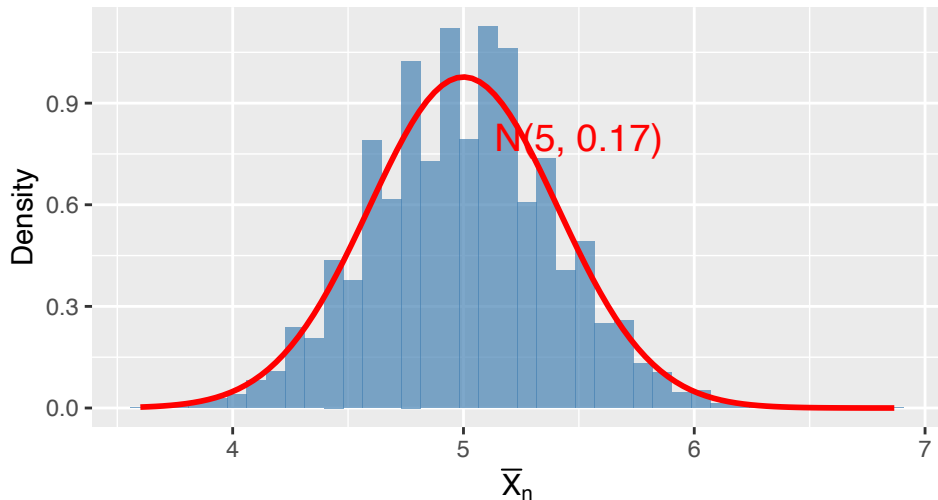
Sample means with data size $n = 5$

Sample means from Poisson(5), $n=5$

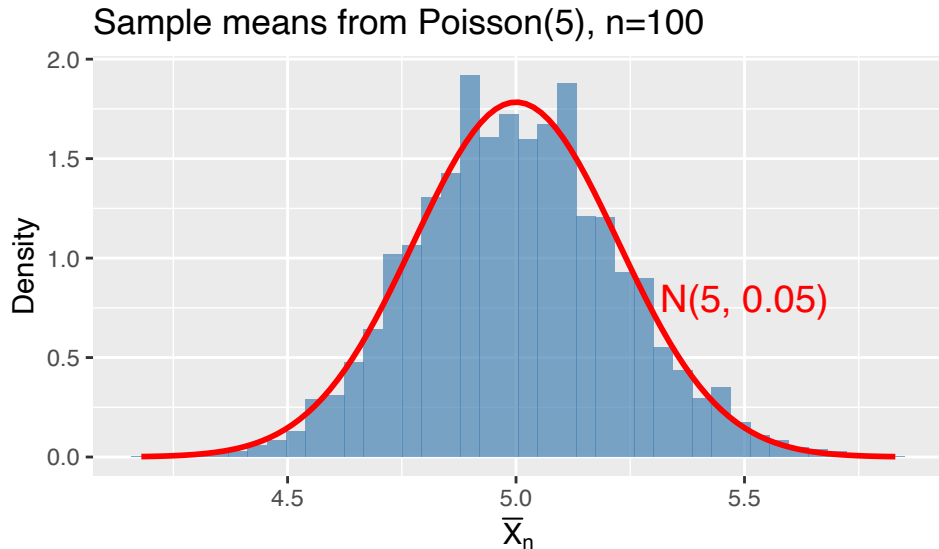


Sample means with data size $n = 30$

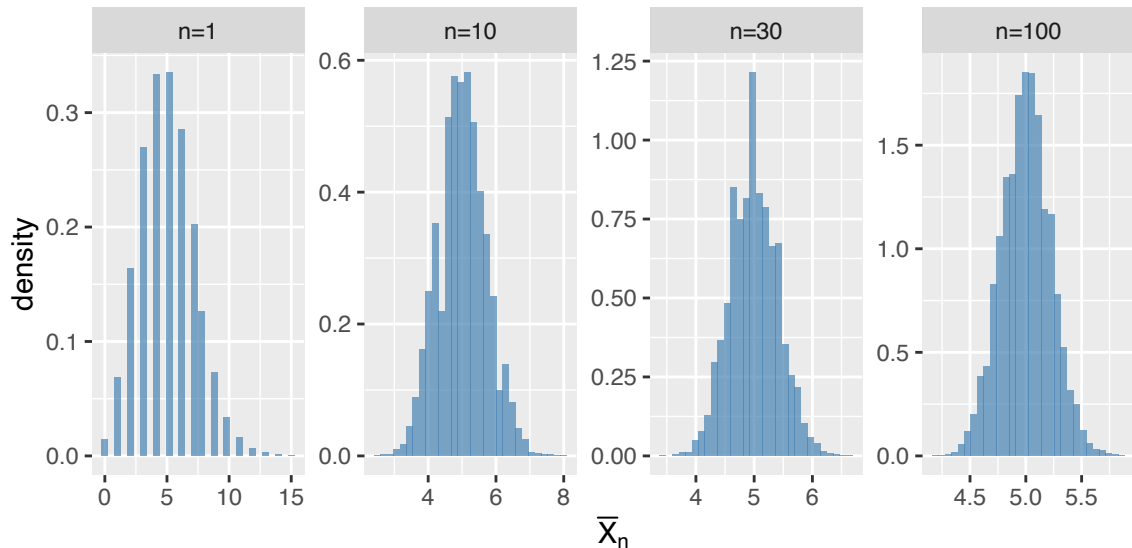
Sample means from Poisson(5), $n=30$



Sample means with $n=100$



Different (data) sample sizes (10k simulations)



Some clarifications

Given $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, i.i.d., $X_i \sim \mathcal{N}(\mu, \sigma^2)$

can you distinguish the following?

$$\mathbb{E}[X_1]$$

vs.

$$\mathbb{E}[\bar{X}_n]$$

$$\mathbb{V}[X_1]$$

vs.

$$\mathbb{V}[\bar{X}_n]$$

Some clarifications

Given $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, i.i.d., $X_i \sim \mathcal{N}(\mu, \sigma^2)$

can you distinguish the following?

$$\mathbb{E}[X_1] = \mu$$

vs.

$$\mathbb{E}[\bar{X}_n] = \mu$$

$$\mathbb{V}[X_1] = \sigma^2$$

vs.

$$\mathbb{V}[\bar{X}_n] = \sigma^2/n$$

What CLT suggests in real world applications...

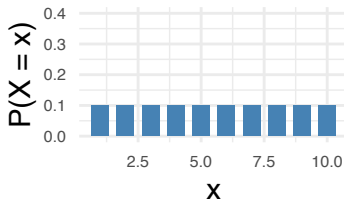
- If your variable of interest is something like sample mean?
- If you have enough sample size?

Can you suggest probability distributions for these?

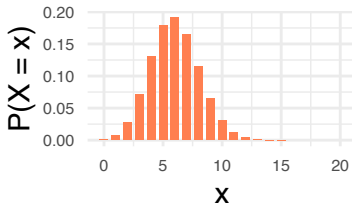
- number of chromosomes
- survival time
- mRNA expression level

Common Discrete Distributions

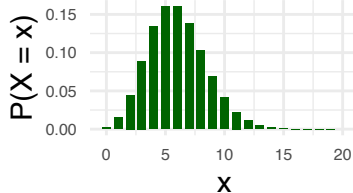
Discrete Uniform(1,10)



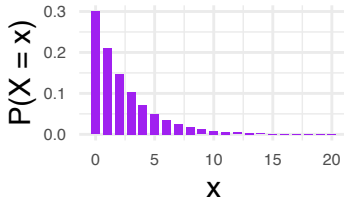
Binomial($n=20$, $p=0.3$)



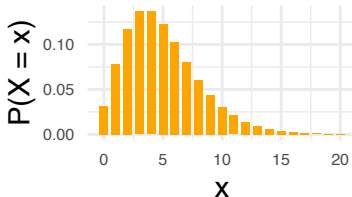
Poisson($\lambda=6$)



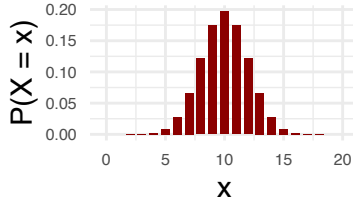
Geometric($p=0.3$)



Negative Binomial
($r=5$, $p=0.5$)

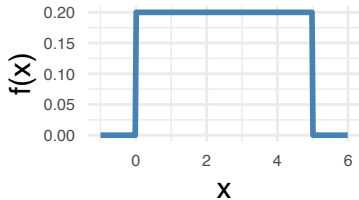


Hypergeometric
($m=50$, $n=50$, $k=20$)

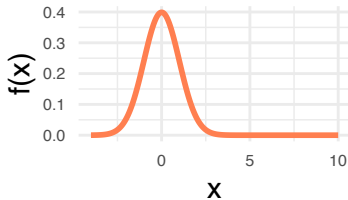


Common Continuous Distributions

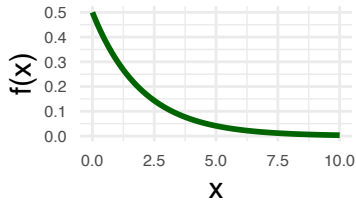
Continuous Uniform
($a=0$, $b=5$)



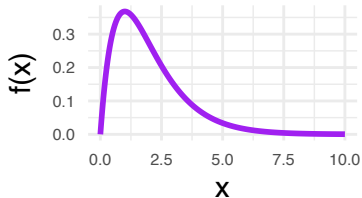
Normal($\mu=0$, $\sigma=1$)



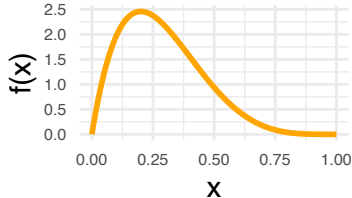
Exponential($\lambda=0.5$)



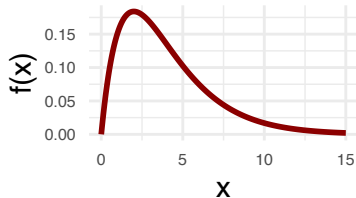
Gamma($\alpha=2$, $\beta=1$)



Beta($\alpha=2$, $\beta=5$)



Chi-squared(df=4)



Discussion

- How do we know **which distribution** works well with our own data?
- Should we use just **one** type of distribution or **many**?
- Can we mix them up in one model?

Pop quiz: What are they?

```
## 1.  
rnorm(...)
```

```
## 2.  
dnorm(...)
```

```
## 3.  
pnorm(...)
```

```
## 4.  
qnorm(...)
```

```
import numpy as np  
import scipy.stats as stats
```

```
## 1.  
np.random.normal(...)
```

```
## 2.  
stats.norm.pdf(...)
```

```
## 3.  
stats.norm.cdf(...)
```

```
## 4.  
stats.norm.ppf(...)
```

Summary

- Random variables are variables that have a probability distribution
- Any statistic of sampled data is a RV, and hence has an associated probability distribution
- The CLT gives us the sampling distribution of the mean of any RV (regardless of its distribution)
- We can use statistical inference to estimate population parameters from a sample

Today's lecture

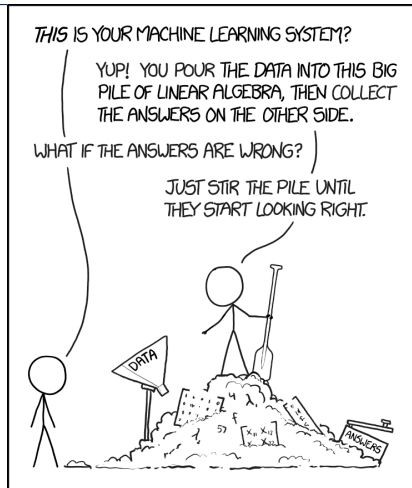
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Why do we use matrix?

Two main reasons:

- ① **Simplify mathematical notation:** Describe many linear models concisely in one line
- ② **Efficient computation:** many programming languages (Python, R, C/C++, Fortran) implement matrix operations efficiently (a library built on decades of optimization)
 - Matrix (linear) algebra is a foundational tool to represent whole space of vectors (many variables).
 - Our long-term goal: We want to think of a multivariate linear regression in terms of “column space.”

AI/ML: It's all about matrix.



Source: xkcd #1838: Machine Learning

Example: A system of linear equations

Solve for a , b , c :

$$a + b + c = 6$$

$$3a - 2b + c = 2$$

$$2a + b - c = 1$$

Source: Matrix Notations

Example: A system of linear equations

Solve for a , b , c :

$$a + b + c = 6$$

$$3a - 2b + c = 2$$

$$2a + b - c = 1$$

Matrix form:

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & -2 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

More compact: $\mathbf{Ax} = \mathbf{b}$

Source: Matrix Notations

Matrix Notation: Scalars, Vectors, Matrices

Scalar

Single number

$$a = 5$$

$$\lambda = 3.14$$

Vector

Ordered collection (bold)

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$n \times 1$ matrix

Matrix

Rectangular array (bold)

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ \vdots & \vdots \\ x_{n,1} & x_{n,2} \end{bmatrix}$$

$n \times p$ matrix

Source: Matrix Notations

Notation conventions

- **(Bold) uppercase** for matrices: $\mathbf{X}$, $\mathbf{A}$
- **Bold lowercase** for vectors: $\mathbf{y}$, $\mathbf{x}$
- Regular lowercase for scalars: a , $x_{i,j}$
- $x_{i,j}$ denotes element in row i , column j of matrix $\mathbf{X}$
- Vectors are typically $n \times 1$ (column vectors) by convention (statistics)
- Vectors are usually $1 \times n$ (row vectors) in many AI/deep learning libraries

Source: Matrix Notations

Example in R: Creating matrices

```
# Create a vector
```

```
y <- c(1, 2, 3, 4, 5)
```

```
y
```

```
## [1] 1 2 3 4 5
```

```
# Create a matrix by binding columns
```

```
X <- cbind(c(1, 1, 1), c(1, 2, 3))
```

```
X
```

```
##      [,1] [,2]
```

```
## [1,]    1    1
```

```
## [2,]    1    2
```

```
## [3,]    1    3
```

Creating matrices in R (continued)

```
# Using matrix() function (fills column by column by  
# default)  
A <- matrix(c(1, 3, 2, 1, -2, 1, 1, 1, -1), nrow = 3, ncol = 3)  
A
```

```
##      [,1] [,2] [,3]  
## [1,]    1    1    1  
## [2,]    3   -2    1  
## [3,]    2    1   -1
```

Creating matrices in R (continued)

```
# Fill row by row instead
```

```
B <- matrix(c(1, 1, 1, 3, -2, 1, 2, 1, -1), nrow = 3, ncol = 3,  
            byrow = TRUE)
```

```
B
```

```
##      [,1] [,2] [,3]  
## [1,]    1    1    1  
## [2,]    3   -2    1  
## [3,]    2    1   -1
```


Pop quiz

Is it row-major or column major?

```
matrix(...)
```

Pop quiz

Is it row-major or column major?

```
matrix(..., byrow = TRUE)
```

Scalar Multiplication

Multiply each element by the scalar:

$$c\mathbf{X} = c \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \end{bmatrix}$$
$$= \begin{bmatrix} c \cdot x_{1,1} & c \cdot x_{1,2} \\ c \cdot x_{2,1} & c \cdot x_{2,2} \end{bmatrix}$$

```
X <- matrix(c(1, 2, 3, 4), 2, 2)
X
```

```
##          [,1] [,2]
## [1,]         1     3
## [2,]         2     4
```

```
# Scalar multiplication
3 * X
```

```
##          [,1] [,2]
## [1,]         3     9
## [2,]         6    12
```

Transpose

Converts rows into columns and columns into rows: $\mathbf{X}^\top$

If $\mathbf{X}$ is $n \times p$, then $\mathbf{X}^\top$ is
 $p \times n$

$$\mathbf{X} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ 3 & 5 \end{bmatrix}_{3 \times 2}$$

$$\mathbf{X}^\top = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}_{2 \times 3}$$

```
X <- matrix(c(1, 2, 3, 3, 4, 5), 3, 2)
X
```

```
##           [,1] [,2]
## [1,]         1     3
## [2,]         2     4
## [3,]         3     5
```

```
# Transpose
t(X)
```

```
##           [,1] [,2] [,3]
## [1,]         1     2     3
## [2,]         3     4     5
```

Matrix Multiplication

Check: Number of columns in first matrix = number of rows in second matrix.

If $\mathbf{A}$ is $n \times p$ and $\mathbf{B}$ is $p \times m$, then
 $\mathbf{AB}$ is $n \times m$.

$$(\mathbf{AB})_{i,j} = \sum_{k=1}^p a_{i,k} b_{k,j}$$

Row i of $\mathbf{A}$ times column j of $\mathbf{B}$

```
A <- matrix(c(1, 2, 3, 4), 2, 2)
B <- matrix(c(5, 6, 7, 8), 2, 2)
```

```
# Matrix multiplication
A %*% B
```

```
##      [,1] [,2]
## [1,]   23  31
## [2,]   34  46
```

Source: Matrix Operations

Matrix Multiplication

Note:

```
# Matrix multiplication
```

```
A %*% B
```

```
##           [,1] [,2]  
## [1,]      23  31  
## [2,]      34  46
```

This is different from:

```
# Element-wise multiplication
```

```
A * B
```

```
##           [,1] [,2]  
## [1,]         5  21  
## [2,]        12  32
```

Source: Matrix Operations

Matrix Multiplication Example

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 3 \cdot 6 & 1 \cdot 7 + 3 \cdot 8 \\ 2 \cdot 5 + 4 \cdot 6 & 2 \cdot 7 + 4 \cdot 8 \end{bmatrix}$$
$$= \begin{bmatrix} 23 & 31 \\ 34 & 46 \end{bmatrix}$$

```
matrix(c(1, 2, 3, 4), 2, 2) %*% matrix(c(5, 6, 7, 8), 2, 2)
```

```
##      [,1] [,2]  
## [1,]   23  31  
## [2,]   34  46
```

Source: Matrix Operations

Identity Matrix

$n \times n$ square matrix with ones on diagonal, zeros elsewhere: $\mathbf{I}_n$

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Property: $\mathbf{XI} = \mathbf{IX} = \mathbf{X}$ (like multiplying by 1)

```
I <- diag(3)
```

```
I
```

```
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    1    0
## [3,]    0    0    1
```


Matrix Inverse

For square matrix $\mathbf{A}$, the inverse $\mathbf{A}^{-1}$ satisfies:

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

Not all matrices have inverses!

```
A <- matrix(c(1, 3, 2, 1, -2, 1, 1, 1, -1), 3, 3, byrow = TRUE)
A_inv <- solve(A)
A_inv
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.07692308 0.3846154 0.53846154
## [2,] 0.15384615 -0.2307692 0.07692308
## [3,] 0.23076923 0.1538462 -0.38461538
```

Verify: Should give identity matrix

For square matrix $\mathbf{A}$, the inverse $\mathbf{A}^{-1}$ satisfies:

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

```
A %*% A_inv
```

```
##           [,1]           [,2]           [,3]
## [1,] 1.000000e+00  0.000000e+00  0.000000e+00
## [2,] 0.000000e+00  1.000000e+00 -1.110223e-16
## [3,] 1.110223e-16 -2.775558e-17  1.000000e+00
```

Solving linear equations with matrix inverse

Recall: $Ax = b$

$$A^{-1}Ax = A^{-1}b$$

$$Ix = A^{-1}b$$

$$x = A^{-1}b$$

```
## solve linear equations
```

```
x <- solve(A, b)
```

```
round(x, 2)
```

```
## [1] 1.77 0.54 1.31
```

```
## another way (less efficient)
```

```
x <- solve(A) %*% b
```

```
round(t(x), 2)
```

```
##      [,1] [,2] [,3]
```

```
## [1,] 1.77 0.54 1.31
```

Summary

- A system of linear equations:

$$A\mathbf{x} = \mathbf{b}$$

- Matrix operations: **multiplication, transpose, inverse**
- Recipe: **The Matrix Cookbook**

Today's lecture

- 1 Statistics Refresher
- 2 Matrix Algebra Primer
- 3 Matrix Algebra in Statistics**

Calculating moments with matrices

Goal: Express mean and variance using matrix operations

Given data vector $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

Sample mean using matrix operations

The sample mean: $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$

Using matrices: Define $\mathbf{1}_n = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$ (vector of ones)

$$\bar{y} = \frac{1}{n} \mathbf{1}_n^\top \mathbf{y}$$

```
y <- c(3, 5, 7, 2, 8)
n <- length(y)
ones <- rep(1, n)
```

```
## Using matrix multiplication
mean.y <- (1/n) * t(ones) %*% y
mean.y

##      [,1]
## [1,]    5
```

Centering data with matrices

Centered data: $y_i - \bar{y}$ for all observations

Matrix form: $\mathbf{y} - \bar{y}\mathbf{1}_n$

```
y <- c(3, 5, 7, 2, 8)
mean.y <- mean(y)
n <- length(y)
```

```
## Centering
ones.n <- rep(1, n)
y.cent <- y - mean.y * ones.n
y.cent
```

```
## [1] -2  0  2 -3  3
```

```
mean(y.cent)
```

```
## [1] 0
```


Sample variance using matrix operations

Sample variance: $s^2 = (n - 1)^{-1} \sum_{i=1}^n (y_i - \bar{y})^2$

Letting (centered data)

$$\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}_n$$

Sample variance using matrix operations

Sample variance: $s^2 = (n - 1)^{-1} \sum_{i=1}^n (y_i - \bar{y})^2$

Letting (centered data)

$$\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}_n$$

$$s^2 = (n - 1)^{-1} \tilde{\mathbf{y}}^\top \tilde{\mathbf{y}}$$

Sample variance using matrix operations

```
y <- c(3, 5, 7, 2, 8)
n <- length(y)
y.tilde <- y - mean(y)
```

```
# using matrix multiplication
var.y <- t(y.tilde) %*% y.tilde
var.y <- var.y/(n - 1)
```

```
# Compare with built-in var
all(var.y == var(y))
```

```
## [1] TRUE
```

Covariance matrix using matrix operations

For two variables X and Y , the covariance matrix is:

$$\Sigma = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$

With centered data matrix $\tilde{\mathbf{X}}$ ($n \times 2$):

$$\Sigma = \frac{1}{n-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$$

Computing 2x2 covariance matrix in R

```
# Two variables
```

```
x1 <- c(3, 5, 7, 2, 8)
```

```
x2 <- c(4, 6, 5, 3, 7)
```

```
n <- length(x1)
```

```
# Create data matrix
```

```
X <- cbind(x1, x2)
```

```
X
```

```
##      x1 x2
```

```
## [1,]  3  4
```

```
## [2,]  5  6
```

```
## [3,]  7  5
```

```
## [4,]  2  3
```

```
## [5,]  8  7
```

```
# Center the data
```

```
X.tilde <- scale(X, scale = FALSE)
```

```
X.tilde
```

```
##      x1 x2
```

```
## [1,] -2 -1
```

```
## [2,]  0  1
```

```
## [3,]  2  0
```

```
## [4,] -3 -2
```

```
## [5,]  3  2
```

```
## attr(,"scaled:center")
```

```
## x1 x2
```

```
##  5  5
```

Computing 2x2 covariance matrix in R (continued)

```
# Covariance matrix
```

```
Sigma <- t(X.tilde) %*% X.tilde
```

```
Sigma <- Sigma/(n - 1)
```

```
Sigma
```

```
##      x1  x2
```

```
## x1 6.5 3.5
```

```
## x2 3.5 2.5
```

```
# Compare with built-in cov
```

```
cov(X)
```

```
##      x1  x2
```

```
## x1 6.5 3.5
```

```
## x2 3.5 2.5
```

Summary: Why use matrix notation?

- ① **Concise representation:** Express complex operations compactly
- ② **Generalization:** Easily extend to multivariate data (multiple variables)
- ③ **Computational efficiency:** Matrix libraries are highly optimized